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Arm Solutions at Lightspeed

Making MIPI DSI power states fit Linux kernel

MIPI DSI in a nutshell

- Data is sent in packets
- Video mode (stream of pixels) or Command mode (draw here) peripherals
- Up to 4 virtual channels
- Several lanes (further defined by the PHY)
  - D-PHY: 1 clock lane, up to 4 data lanes, differential pairs
  - C-PHY: up to 3 lanes, embedded clock, 3 wires
- Can be split into several Sub-Links (by distributing lanes)
- Optional bidirectional lane
- RGB data, can support YCbCr, DSC, HDR, etc.
- LP (slow) or HS (fast) transfers

MIPI DSI in a nutshell

- Three bus states:
  - Off
  - Clock on
  - Video on
- Usually one peripheral on a bus
  - Can have more than one

Mapping to DRM

- peripheral::pre_enable
  - host::pre_enable
- encoder::enable
  - host::enable
    - peripheral::enable

enable video stream
power on DSI bus, switch to HS mode
send DSI commands

SUN4i: no DSI commands after video stream!
How to shutdown the panel?

Mapping to DRM, try2

- host::mode_set	power on DSI bus
- peripheral::pre_enable	send DSI commands
- host::pre_enable	switch to HS mode
- encoder::enable	enable video stream
- host::enable	
- peripheral::enable	
- peripheral::disable	shutdown the panel?
- host::disable	
- encoder::disable	disable video stream
- host::post_disable	shutdown DSI bus???
- peripheral::post_disable	shutdown the panel?

Special cases (Parade PS8640)

Mapping to DRM, pre_enable_prev_first

- host::mode_set	
- host::pre_enable	power on DSI bus
- peripheral::pre_enable	send DSI commands
- encoder::enable	enable video stream
- host::enable	switch to HS mode
- peripheral::enable	
- peripheral::disable	
- host::disable	
- encoder::disable	disable video stream
- peripheral::post_disable	shutdown the panel
- host::post_disable	shutdown DSI bus

Does peripheral use pre_enable_prev_first???
Multiple devices on a bus?

DSI bus callbacks

```
host_pre_enable()
{
    for_each_dsi_device()
        dsi->power_on();

    power_on_bus();

    for_each_dsi_device()
        dsi->lp_commands();
}
host_enable()
{
    for_each_dsi_device()
        dsi->hs_commands();
}
```

Mapping to DRM, DSI bus callbacks

- host::mode_set	
- host::pre_enable	power on DSI bus make peripherals send commands
- encoder::enable	enable video stream
- host::enable	switch to HS mode
- host::disable	
- encoder::disable	disable video stream
- host::post_disable	make peripherals to shutdown shutdown DSI bus

DSI bus callbacks, Pro et Contra

- No feature negotiation
- Clean bus state separation
- Multiple panels handled logically!

- Rewrite all hosts
- Rewrite all clients
- Multiple panels enabled simultaneously :-(
- Doesn’t use drm_panel funcs anymore

- ULPS?

Mapping to DRM, Runtime PM

- peripheral::pre_enable	pm_runtime_get_sync(dsi_host) send DSI commands
- host::pre_enable	
- encoder::enable	enable video stream
- host::enable	switch to HS mode
- peripheral::enable	
- peripheral::disable	shutdown the panel pm_runtime_put(dsi_host)
- host::disable	
- encoder::disable	disable video stream
- host::post_disable	
- peripheral::post_disable	

Runtime PM, Pro et Contra

- No feature negotiation
- Clean bus state separation

- Rewrite all hosts and clients
- Multiple peripherals case not solved

- ULPS?

Thank You!

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